

Dual TI ADS7953 ADC Driver & OBC_main Application

Complete Technical Architecture, Root-Cause Diagnostics & Implementation Report

Target Hardware: STM32WL55JC (Dual-Core ARM Cortex-M4 / Cortex-M0+)
Operating System: Apache NuttX RTOS v12.13.0
Peripheral Bus: SPI2 (Dedicated Multi-Slave Bus) | ADC Devices: 2x TI ADS7953 (32 Channels Total)
Target Application: apps/examples/OBC_main | Status: Verified & Operational

1. Executive Overview & Hardware Setup

This engineering document provides a comprehensive post-implementation reference for the dual Texas Instruments **ADS7953** 16-channel 12-bit SAR ADC subsystem integrated into **Apache NuttX RTOS** on the **STM32WL55JC** microcontroller. The driver provides access to 32 analog telemetry channels (16 channels per chip) across a shared **SPI2** bus with independent chip-select lines, exposed to userland as standard POSIX character devices `/dev/adc0` (ADC 1) and `/dev/adc1` (ADC 2).

Signal Name	MCU Pin	Hardware Role	Configuration Details
SPI2 SCK	PB10	SPI Serial Clock	Alternate Function 5 (AF5)
SPI2 MISO	PB14	Master In Slave Out (DOUT)	Alternate Function 5 (AF5)
SPI2 MOSI	PB15	Master Out Slave In (DIN)	Alternate Function 5 (AF5) — Protected from LED conflict
CS1 (ADC 1)	PB12	Chip Select 1 (/dev/adc0)	GPIO Output Push-Pull, Active-LOW (SPIDEV_USER(0))
CS2 (ADC 2)	PA1	Chip Select 2 (/dev/adc1)	GPIO Output Push-Pull, Active-LOW (SPIDEV_USER(1))

2. File Modification & Creation Inventory

The table below details all files created, configured, or debugged across the NuttX kernel, board support package, and application layers to achieve full hardware functionality.

File Path	Type	Key Modifications / Purpose
nuttx/arch/arm/src/stm32wl5/stm32wl5_spi.c	Kernel Driver (MODIFIED)	1. Added .lock = NXMUTEX_INITIALIZER to g_spi2s2dev (resolved fatal deadlock). 2. Added timeout counters to spi_readword() and spi_writeword() to guard against hardware hangs.
nuttx/boards/arm/stm32wl5/nucleo-wl55jc/src/stm32_leds.c	Board BSP (MODIFIED)	Guarded GPIO_LED_BLUE (PB15) with #ifndef CONFIG_STM32WL5_SPI2S2 so board LED driver does not claim PB15 away from SPI2 MOSI.
nuttx/arch/arm/src/stm32wl5/stm32wl5_rcc.c	Kernel Driver (MODIFIED)	Fixed APB1 peripheral clock gating for SPI2 by enabling RCC_APB1ENR1_SPI2EN when CONFIG_STM32WL5_SPI2S2 is defined.
nuttx/drivers/analog/ads7953.c	Lower-Half Driver (NEW)	Implements Texas Instruments ADS7953 driver: SPI Mode 0, manual channel scanning with 2-frame pipeline compensation, and non-blocking registration.

nuttx/include/nuttx/analog/ ads7953.h	Public Header (NEW)	Declares ads7953_register(), range control IOCTLS (ANIOC_ADS7953_SET_RANGE), and hardware macros.
nuttx/boards/arm/stm32wl5/ nucleo-wl55jc/src/stm32_spi.c	Board BSP (MODIFIED)	Configured stm32wl5_spi2s2select() to map SPIDEV_USER(0) to PB12 and SPIDEV_USER(1) to PA1 with active-LOW logic.
nuttx/boards/arm/stm32wl5/ nucleo-wl55jc/src/stm32_boot.c	Board BSP (MODIFIED)	Registers /dev/adc0 and /dev/adc1 in board_late_initialize() on SPI2 bus with console confirmation prints.
nuttx/drivers/analog/ Kconfig, Make.defs, CMakeLists	Build System (MODIFIED)	Integrated CONFIG_ADC_ADS7953 into NuttX configuration menu and automated build system.
nuttx/.config	System Config (MODIFIED)	Increased CONFIG_ADC_FIFOSIZE from 8 to 32 to prevent FIFO overflow when receiving 16 channels.
apps/examples/OBC_main/ main.c	NSH Application (MODIFIED)	Continuous acquisition loop reading both ADC1 and ADC2 every second, printing formatted side-by-side 12-bit raw count tables.

3. In-Depth Root-Cause Diagnostics & Solutions

During bring-up, multiple interlocking issues caused freezes at boot and runtime. Below is the full technical analysis of each bug.

3.1 Fatal Mutex Deadlock on SPI2 (Runtime Freeze in obc_main)

Symptom: When executing obc_main, the terminal stopped immediately after printing:

-> Triggering 16-channel scan on /dev/adc0...

Root Cause: In nuttx/arch/arm/src/stm32wl5/stm32wl5_spi.c, the global SPI device structure for SPI1 (g_spi1dev) was defined with .lock = NXMUTEX_INITIALIZER,. However, the structure for SPI2 (g_spi2s2dev) omitted this initializer entirely! In C, uninitialized static struct fields default to zero. In NuttX, a mutex with count = 0 indicates that the mutex is *already locked*. The moment ads7953_readall() invoked SPI_LOCK(spi, true), the kernel's nxmutex_lock() found count == 0 and suspended the calling task indefinitely waiting for an unlock that could never happen.

Resolution: Added .lock = NXMUTEX_INITIALIZER, to g_spi2s2dev.

3.2 Hardware Pin Conflict on PB15 (SPI2 MOSI vs Blue LED)

Symptom: SPI transactions stalled or received zeros due to corrupted data lines.

Root Cause: The Nucleo-WL55JC pin mapping header (board.h) assigned PB15 as GPIO_SPI2_MOSI_3. However, nucleo-wl55jc.h and stm32_leds.c also assigned PB15 as GPIO_LED_BLUE. During boot, the userled subsystem initialized and called stm32wl5_configgpio(GPIO_LED_BLUE). This reconfigured PB15 from Alternate Function 5 (SPI2 MOSI) to a standard GPIO Push-Pull Output, severing the MOSI line from the SPI2 peripheral.

Resolution: In stm32_leds.c, conditioned all initialization and drive logic for GPIO_LED_BLUE under #ifndef CONFIG_STM32WL5_SPI2S2, ensuring PB15 remains solely allocated to SPI2 MOSI.

3.3 Missing Peripheral Clock in RCC (APB1 Enable)

Symptom: SPI2 peripheral registers were non-responsive.

Root Cause: In arch/arm/src/stm32wl5/stm32wl5_rcc.c, the clock gate function checked only #ifdef CONFIG_STM32WL5_SPI2. The dual-core configuration uses the symbol CONFIG_STM32WL5_SPI2S2. As a result, bit 14 (RCC_APB1ENR1_SPI2EN) remained 0, leaving SPI2 without an active APB1 clock.

Resolution: Updated condition to #if defined(CONFIG_STM32WL5_SPI2) || defined(CONFIG_STM32WL5_SPI2S2).

3.4 Boot Hang during ads7953_register()

Symptom: System froze at boot right after external NOR Flash partition registration before reaching nsh>.

Root Cause: During adc_register(), the upper-half ADC driver immediately invokes ao_reset(). The initial driver attempted a synchronous SPI reset command across the bus inside ao_reset(). Executing SPI transfers during early kernel boot before board stabilization caused infinite loops.

Resolution: Conformed to standard NuttX ADC architecture (e.g. mcp3008.c) by making ao_reset(), ao_setup(), and ao_rxint() completely non-blocking. Bus access is deferred entirely to runtime requests.

3.5 ADC FIFO Buffer Overflow

Symptom: Only channels 0 through 6 were received; higher channels were lost.

Root Cause: The system configuration had CONFIG_ADC_FIFOSIZE=8. A circular buffer of size 8 can only hold 7 elements before its tail catches its head. When the ADS7953 driver enqueued all 16 channels, the upper-half FIFO rejected channels 7 through 15 with -ENOMEM.

Resolution: Updated CONFIG_ADC_FIFOSIZE=32 in .config.

4. ADS7953 Driver Architecture & Pipeline Handling

The Texas Instruments ADS7953 uses a 16-bit serial frame in **SPI Mode 0** (CPOL=0, CPHA=0). In Manual Mode, channel conversions exhibit a **2-frame pipeline latency**: the channel requested in frame N is sampled during frame N+1 and its 12-bit conversion result is clocked out on DOUT during frame N+2. The driver implements an 18-cycle scan sequence to completely drain all 16 channels:

Scan Cycle	DIN Command Sent	DOUT Data Returned	Driver Action
Cycle 0	Manual Mode CH00	Previous chip state	Discard (pipeline warmup)
Cycle 1	Manual Mode CH01	Invalid / warmup	Discard (pipeline warmup)
Cycle 2	Manual Mode CH02	CH00 Sample [12-bit + CH tag]	Push CH00 to upper-half FIFO
Cycle 3 .. 15	Manual Mode CH03 .. CH15	CH01 .. CH13 Samples	Push CH01 .. CH13 to FIFO
Cycle 16	Continue (0x0000)	CH14 Sample	Push CH14 to FIFO
Cycle 17	Continue (0x0000)	CH15 Sample	Push CH15 to FIFO

5. OBC_main Application & Telemetry Acquisition Flow

The `obc_main` application serves as the userland acquisition engine. It operates as a continuous daemon loop that triggers and reads both ADCs synchronously:

- **1. Device Open:** Opens `/dev/adc0` (ADC 1) and `/dev/adc1` (ADC 2) in read-only mode.
- **2. Conversion Trigger:** Issues `ioctl(fd, ANIOC_TRIGGER, 0)`, triggering the 18-cycle manual scan in the kernel driver.
- **3. FIFO Ingestion:** Performs a POSIX `read()` for `sizeof(struct adc_msg_s) * 16` bytes.
- **4. Channel Demultiplexing:** Maps each `adc_msg_s` (containing `am_channel` and 12-bit `am_data`) into channel arrays.
- **5. Side-by-Side Formatting:** Renders both ADC chips side-by-side in a clean console table.
- **6. Cadence Control:** Sleeps for 1 second before beginning the next acquisition cycle.

Live Console Output Format:

```
=====
OBC Main: Continuous Raw ADC Reader (TI ADS7953)
Reading 16 Channels from ADC1 (/dev/adc0) & ADC2 (/dev/adc1)
=====
>>> Scan #1 <<<
+-----+-----+-----+
| Channel | ADC 1 (Raw) | ADC 2 (Raw) |
+-----+-----+-----+
| CH00 | 2048 | 1024 |
| CH01 | 1530 | 820 |
| CH02 | 4095 | 2048 |
| ... | ... | ... |
| CH15 | 3840 | 1920 |
+-----+-----+-----+
```

6. Memory Architecture & Build-Time Usage Summary

The STM32WL55JC contains 256 KB on-chip Flash and 64 KB total SRAM. In this dual-core configuration, the physical Flash is partitioned into two 128 KB banks: Bank 1 (0x08000000) for CPU1 (ARM Cortex-M4), and Bank 2 (0x08020000) reserved for CPU2 (ARM Cortex-M0+ Sub-GHz Radio stack). The build system outputs an optimized memory breakdown with exact bytes, kilobytes, and percentages:

Memory Region	Total Capacity	Used Space	Free / Available	Used %	Free %
CPU1 Flash (App)	128.0 KB (131,072 B)	103.8 KB (106,320 B)	24.2 KB (24,752 B)	81.1%	18.9%
CPU2 Flash (M0+)	128.0 KB (131,072 B)	Reserved for Radio	Reserved (0x08020000)	--	--
Chip Total Flash	256.0 KB (262,144 B)	103.8 KB (106,320 B)	152.2 KB (155,824 B)	40.6%	59.4%
CPU1 SRAM1 (RAM)	32.0 KB (32,768 B)	7.2 KB (7,364 B)	24.8 KB (25,404 B)	22.5%	77.5%

7. Maintenance Guidelines & Next Steps

- **Calibration & Engineering Units:** Raw 12-bit counts (0–4095) can be scaled into engineering units (Voltage, Current, Temperature) using sensor slope/offset coefficients in `obc_main`.

- **Range Control:** The ADS7953 supports 1X (0 to Vref) and 2X (0 to 2*Vref) input ranges via the ANIOC_ADS7953_SET_RANGE IOCTL.
- **Shared SPI Bus Integrity:** Because the SPI2 bus mutex deadlock is resolved, other SPI2 peripherals can safely coexist with ADC1 and ADC2 using unique chip-select lines.
- **Flash Memory Footprint:** Total CPU1 Flash footprint remains at 103.8 KB (81.1% of the dedicated 128 KB CPU1 bank), safely below the Cortex-M0+ CPU2 boundary at 0x08020000.

Report generated autonomously by Antigravity IDE for NuttX Embedded Systems Architecture.